



Tech_Store



Project Overview

Tech_Store - Một ứng dụng E-commerce hoàn chỉnh xây dựng với MERN Stack

Timeline: 30 ngày | **Status:**  In Progress | **Progress:** 0/30 tasks



Project Goals

- Xây dựng full-stack e-commerce platform
- Áp dụng các best practices trong MERN development
- Học và thực hành deployment workflow
- Tạo portfolio project chất lượng cao



Tech Stack

Frontend

-  React.js
-  Tailwind CSS / Material-UI
-  Redux / Context API
-  Responsive Design

Backend

-  Node.js & Express
-  MongoDB & Mongoose
-  JWT Authentication
-  RESTful API

DevOps & Tools

-  Vercel / Render
-  Git & GitHub
-  Testing Libraries
-  Analytics



Progress Tracker

Current Phase: Phase 1 - Foundation & Setup

Next Milestone: Complete Authentication System

Completed: 0/30 tasks (0%)

Phase 1: Foundation & Setup (Days 1-7)

- ☐ Day 1 — Khởi tạo dự án
- ☐ Day 2 — Thiết kế Database
- ☐ Day 3 — Authentication Backend
- ☐ Day 4 — Authentication Frontend
- ☐ Day 5 — Product Backend
- ☐ Day 6 — Product Frontend
- ☐ Day 7 — Category + Search

Phase 2: Core Features (Days 8-15)

- ☐ Day 8 — Cart System
- ☐ Day 9 — Cart UI
- ☐ Day 10 — Checkout
- ☐ Day 11 — Order History
- ☐ Day 12 — Admin Authentication
- ☐ Day 13 — Admin Product CRUD
- ☐ Day 14 — Admin Category CRUD
- ☐ Day 15 — Dashboard Statistics

Phase 3: Advanced Features (Days 16-23)

- ☐ Day 16 — Order Management
- ☐ Day 17 — User Management
- ☐ Day 18 — Wishlist
- ☐ Day 19 — Review System
- ☐ Day 20 — Responsive UI
- ☐ Day 21 — Loading + Error Handling

☐ Day 22 — Validation

☐ Day 23 — Security

Phase 4: Optimization & Deployment (Days 24-30)

☐ Day 24 — Pagination

☐ Day 25 — SEO cơ bản

☐ Day 26 — Testing

☐ Day 27 — Bug Fix

☐ Day 28 — Deploy Backend

☐ Day 29 — Deploy Frontend

☐ Day 30 — Hoàn thiện Portfolio

Project Resources

Documentation

[Tóm tắt dự án](#)

Quick Links

-  [GitHub Repository](#)
-  [Live Demo](#)
-  [API Documentation](#)



Development Timeline

Week 1: Foundation

[Day 1 — Khởi
tạo dự án](#)

[Day 2 — Thiết
kế Database](#)

[Day 3 —
Authentication
Backend \(1\)](#)

Week 2: Core Features

[Day 8 — Cart
System \(1\)](#)

[Day 9 — Cart
UI \(1\)](#)

[Day 10 —
Checkout \(1\)](#)

Week 3: Advanced Features

[Day 15 —
Dashboard
Statistics \(1\)](#)

[Day 16 —
Order
Management
\(1\)](#)

Week 4: Polish & Deploy

[Day 22 —
Validation \(1\)](#)

[Day 23 —
Security \(1\)](#)

[Day 24 —
Pagination \(1\)](#)

<u>Day 4 — Authentication Frontend (1)</u>	<u>Day 11 — Order History (1)</u>	<u>Day 17 — User Management (1)</u>	<u>Day 25 — SEO cơ bản (1)</u>
<u>Day 5 — Product Backend (1)</u>	<u>Day 12 — Admin Authentication (1)</u>	<u>Day 18 — Wishlist (1)</u>	<u>Day 26 — Testing (1)</u>
<u>Day 6 — Product Frontend (1)</u>	<u>Day 13 — Admin Product CRUD (1)</u>	<u>Day 19 — Review System (1)</u>	<u>Day 27 — Bug Fix (1)</u>
<u>Day 7 — Category + Search (1)</u>	<u>Day 14 — Admin Category CRUD (1)</u>	<u>Day 20 — Responsive UI (1)</u>	<u>Day 28 — Deploy Backend (1)</u>
		<u>Day 21 — Loading + Error Handling (1)</u>	<u>Day 29 — Deploy Frontend (1)</u>
			<u>Day 30 — Hoàn thiện Portfolio (1)</u>



Key Features



▼ User Features

- Product browsing & search
- Shopping cart management
- Secure checkout process
- Order tracking
- Wishlist functionality
- Product reviews & ratings
- User authentication & profile



▼ Admin Features

- Product CRUD operations
- Category management
- Order management

- User management
- Dashboard with analytics
- Inventory tracking

▼ Security Features

- JWT authentication
- Password hashing
- Input validation
- XSS protection
- CORS configuration
- Rate limiting



Notes & Best Practices

Development Tips:

- Commit code regularly với meaningful messages
- Test từng feature trước khi chuyển sang feature mới
- Document API endpoints trong quá trình phát triển
- Keep components small and reusable
- Follow naming conventions consistently

Common Pitfalls to Avoid:

- Not handling loading states
- Missing error handling
- Poor database schema design
- Inconsistent UI/UX
- Skipping input validation

 Task Tracker

Các collection trong DATABASE

DATA MODELLING